

2 The photograph shows a cane toad, *Rhinella marina*.



(Source: © Natalia Kuzmina/Alamy Stock Photo)

This amphibian is an invasive species in northern Australia.

Pesticides can kill these amphibians quickly.

The outer layer of amphibian skin is only 1 or 2 cells thick. This compares with the skin of mammals, which is 20 to 30 cells thick.

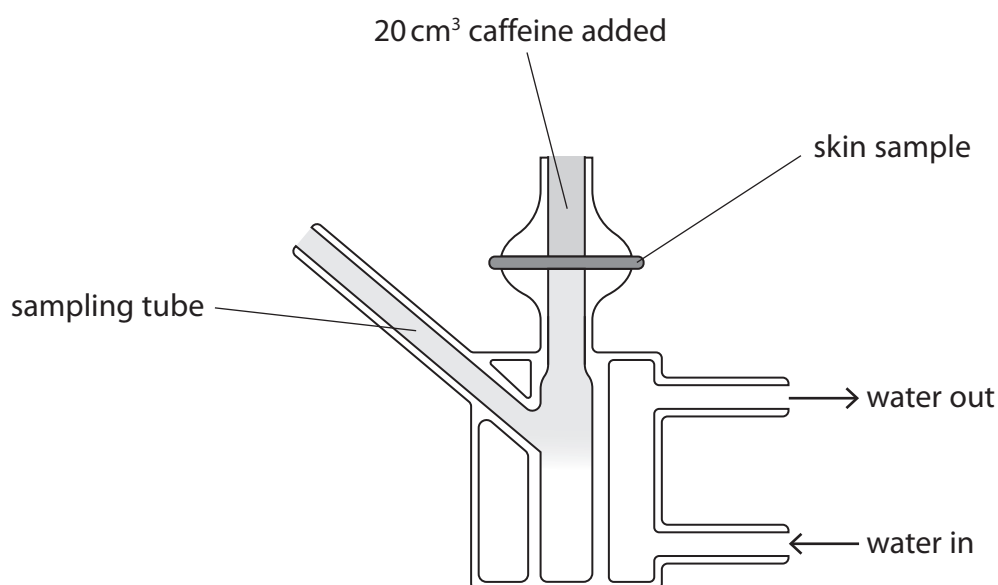
Caffeine is a chemical with similar properties to pesticides.

A scientist investigated the diffusion of caffeine across skin samples from cane toads.

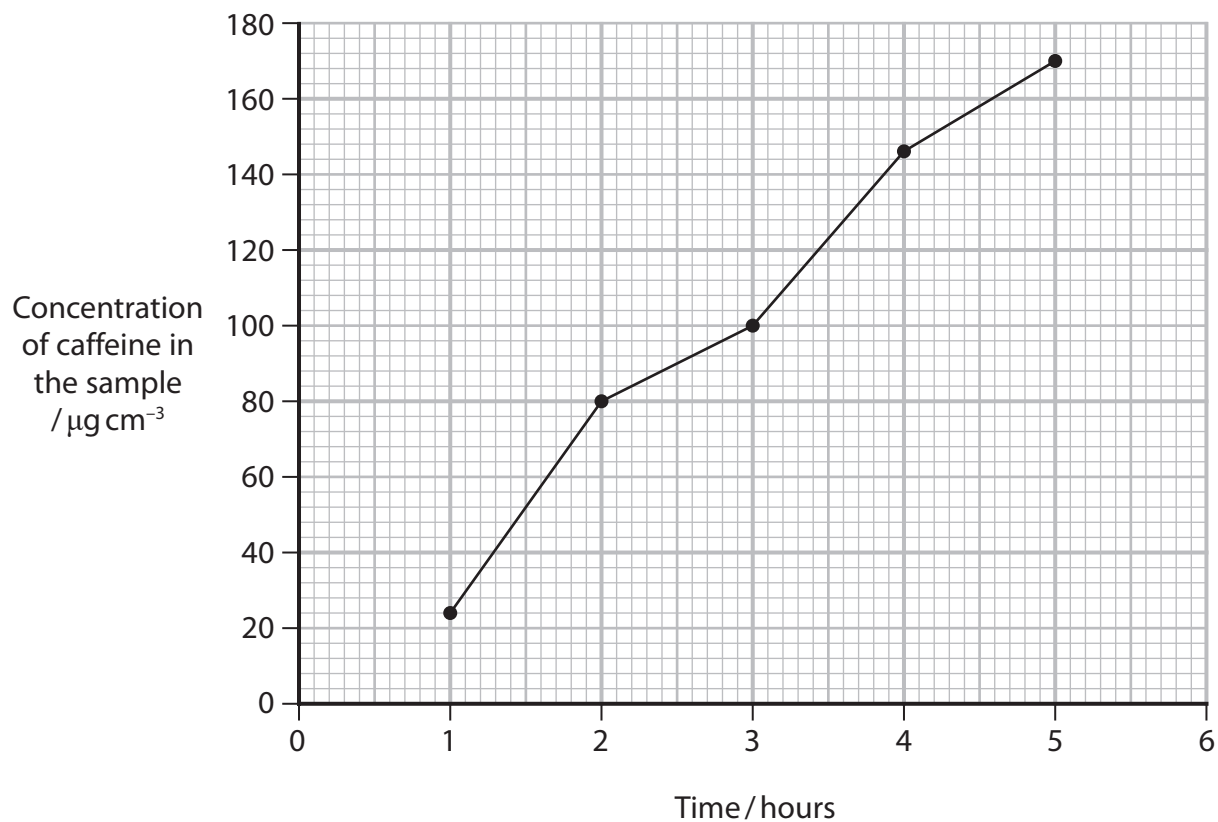
Caffeine solution with a concentration of 200 mg dm^{-3} was added to the top chamber of a diffusion cell.

Samples of solution from the bottom chamber were removed every hour and the concentration of caffeine measured.

The diagram shows a diffusion cell used to measure the rate of diffusion across a skin sample.



The graph shows the results of this investigation.



- (a) (i) Describe how 20 cm^3 of caffeine solution with a concentration of 200 mg dm^{-3} could be prepared.

(1)

- (ii) Calculate the rate of diffusion per minute between 1 and 2 hours.

(1)

..... $\mu\text{g cm}^{-3}\text{ min}^{-1}$

(iii) The rate of diffusion between 4 and 5 hours is $0.40 \mu\text{g cm}^{-3} \text{ min}^{-1}$.

Calculate the percentage decrease in the rate of diffusion between 1 and 2 hours and between 4 and 5 hours.

(1)

..... %

(iv) Caffeine is a polar molecule.

Explain the difference in the rate of diffusion between 1 and 2 hours compared with the rate of diffusion between 4 and 5 hours.

(2)

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(b) (i) State **one** abiotic and **one** biotic variable that could affect this investigation.

(2)

Abiotic variable

Biotic variable

(ii) Choose **one** of the variables you have identified in (b)(i).

Describe how this variable could be controlled and the effect it could have on the results if it is not controlled.

(2)

Variable

Describe how this variable is controlled.

Describe the effect it could have on the results if it is not controlled.

(Total for Question 2 = 9 marks)